Mandatory APS ELN Practice Assignment

Deadline: November 16, 2026 · Textbook & Data Packages: `/MasteringELN/` and `/MasteringELN/assignment/` on the CSMIC Lab Server (`Z:`).
Every student must complete both practice projects inside their own ELN account by the deadline. Both projects will be audited automatically via the ELN MCP server.

Step 1: Download the Raw Data Packages from the Lab Server

Open `/MasteringELN/assignment/` on the CSMIC server (or `Z:\MasteringELN\data\`) and unpack both archives into separate local folders:

1. **Study A (`261015A`, BN/DLC tribological coating, 11 raw files):** `Study-A-261015A-raw.zip` (contains `.depo.json`, `.run.json`, `.tsv`, three step-height `.fibps` scans, `.mit` nanoindentation, `.zrw` friction test, three wear-track `.fibps` scans, and `checksums.md5`).
2. **Study B (`261014B`, Co/C multilayer X-ray mirror, 7 raw files):** `Study-B-261014B-raw.zip` (contains 3 deposition files, 3 plasma-cleaning files, the raw reflectivity scan `XRR-1_Co-C(261014B).xrdml`, and `checksums.md5`). The `.xrcx` fit file is omitted so you produce it yourself in Task 2.

Step 2: Create Your Personal Practice Group and Two Practice Projects

Log into **APS ELN** (`ELabNotebook3.exe`) with your own laboratory account. Never record practice data inside shared production projects:

1. In the left **Projects** panel, choose **Project > New Group** (or right-click an empty area of the Projects panel and choose **New Group...**). Enter the exact group name `Practice` and click **OK** (*Chapter 5*).
2. Right-click your `Practice` group and choose **New Project...**. Create your first project with the exact title `Study A Practice` and click **Save**.
3. Right-click your `Practice` group again and choose **New Project...**. Create your second project with the exact title `Study B Practice` and click **Save**.

Step 3: Complete Task 1 (Desktop GUI) and Task 2 (AI Agent + XRC MCP)

Task 1: Study A (`261015A`) in the Desktop GUI (inside `Study A Practice`)

Read **Chapters 1–4** of *Mastering ELN* and follow the four steps of **Chapter 4** inside `Study A Practice`:

1. **Build the 6-item tree first:** `Deposition 261015A` (Procedure) → `261015A` (Specimen) → child actions `Thickness 261015A`, `Indentation 261015A`, and `Friction test 261015A` (with child action `Wear track 261015A` under `Friction test 261015A`).
2. **Attach all 11 raw files** from `Study-A-261015A-raw.zip` to their matching items (*Table 4-1*).
3. **Extract all 23 `Reader` values** using **Extract Values** and **Extract values from 3 files...** (12 on `Deposition 261015A`, 3 on `Thickness 261015A`, 2 on `Indentation 261015A`, 3 on `Friction test 261015A`, and 3 on `Wear track 261015A`).
4. **Compute the 1 `Derived` wear rate** on `Friction test 261015A` with **Add > Derived Value...** (`wear_rate_reciprocating`, K = 3.05×10^{-7} $mm^3/(N \cdot mm)$), then right-click `Deposition 261015A` and run **Check Value Sources...** (24 unchanged).

Task 2: Study B (`261014B`) with Your AI Agent and `XRC_MCP` (inside `Study B Practice`)

Read **Chapters 15–17** of *Mastering ELN*. Unpack the 7 raw files from `Study-B-261014B-raw.zip` into your configured `--upload-root` directory and use your AI agent (`ELNMCPServer.exe` + `XRC_MCP.exe`) inside `Study B Practice`:

1. **Build the tree and extract 18 deposition/cleaning values (*Chapter 17, Part 1*):** Have the agent create folder `261014B` inside `Study B Practice`, create `Deposition 261014B` (Proc), `Cleaning 261014B` (Action), `261014B` (Specimen), and `XRR-1 261014B` (Action), upload the 7 raw files, and store the 18 `Reader` values (14 on `Deposition 261014B` and 4 on `Cleaning 261014B`).
2. **Fit `XRR-1_Co-C(261014B).xrdml` via `XRC_MCP` (*Chapter 17, Part 2*):** Have the agent fit the curve (`SodaLimeGlass` / [C/Co] × 20 / surface C layer, polynomial order 1) in `XRC_MCP.exe` and save the accepted `.xrcx` file using `curves.job_id` so `[FITRESULT]` and χ^2 are written into the archive.
3. **Upload the `.xrcx` fit and extract the 4 fit values:** Have the agent create child action `XRR fit 261014B` under `XRR-1 261014B`, upload the saved `.xrcx` file, and store the 4 `Reader` values (`Period`, `Co Thickness`, `C Thickness`, and `Chi2`) via `aps.xrcx`.

Warning (No Manual Values): Every stored number in both projects is checked for `Reader` or `Derived` provenance. Typing a number by hand (`Manual` source) or saving an `.xrcx` without `curves.job_id` (missing `[FITRESULT]` / `Chi2`) fails the automated audit.

Step 4: Verification Summary Before November 16, 2026 (Chapter 20)

Project in `Practice` Group	Items	Attached Files	Stored Values	Required Checks Before Deadline
`Study A Practice` (`261015A`)	6 items	11 raw files	23 `Reader` + 1 `Derived` (24)	**Data Quality Check...** (0 issues) & **Check Value Sources...** (24 unchanged)
`Study B Practice` (`261014B`)	5 items	7 raw + 1 `.xrcx` (8)	22 `Reader` (14 + 4 + 4)	`check_storage` (8 ok), `check_value_drift` (22 unchanged), `report_value_issues` (0)